

# **PREFABRICATON**

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## Definition :

- Prefabrication is the practice of assembling components of a structure in a factory or other manufacturing site, and transporting complete assemblies or sub-assemblies to the construction site where the structure is to be located.

## Meaning :

- The assembly of buildings or their components at a location other than the building site.
- The method controls construction costs by economizing on time, wages, and materials.
- Prefabricated units may include doors, stairs, window, walls, wall panels, floor panels, roof trusses, room-sized components, and even entire buildings.
- The term is used to distinguish this process from the more conventional construction practice of transporting the basic material to the construction site where all assembly is carried out.

# PREFABRICATION

Prefabricated building is the completely assembled and erected building, of which the structural parts consist of prefabricated individual units or assemblies using ordinary or controlled materials.



# PREFABRICATION

Prefabricated construction is a new technique and is desirable for large scale housing programmes, yet this has to take a firm hold in the country.





## **AIMS OF PREFABRICATION CONSTRUCTION**

- ☐ Prefabrication is used to effect economy in cost.
- ☐ Components manufactured under controlled conditions.
- ☐ The speed of construction is increased since no curing period is necessary.
- ☐ Prefabrication helps in the use of locally available materials with required characteristics like light-weight; easy workability, thermal insulation, non-combustibility, etc.

## **ADVANTAGES OF PREFABRICATION**

- High quality product
- Labor related savings
- Savings in time
- Overall efficiency is greatly increased
- Mass production is easier and quick
- Protected and controlled production environment
- Potential for lower production costs and other cost savings
- Independence of climatic conditions
- The disruption of traffic is avoided
- Ensures high degree of Safety

## **CHARACTERISTICS ARE TO BE CONSIDERED**

- Easy availability;
- Light weight for easy handling and transport, and to economies on sections and sizes of foundations;
- Thermal insulation property;
- Easy workability;
- Durability in all weather conditions;
- Non-combustibility;
- Economy in cost, and
- Sound insulation.



## Disadvantages :

- Careful handling of prefabricated components such as concrete panels or steel and glass panels is required.
- Attention has to be paid to the strength and corrosion-resistance of the joining of prefabricated sections to avoid failure of the joint.
- Similarly, leaks can form at joints in prefabricated components.
- Transportation costs may be higher for voluminous prefabricated sections than for the materials of which they are made, which can often be packed more compactly.
- Large prefabricated sections require heavy-duty cranes and precision measurement and handling to place in position.



## **MATERIALS TO BE USED IN PREFABRICATION SYSTEM**

- Concrete,
- Steel,
- Treated wood,
- Aluminum,
- Cellular concrete,
- Light weight concrete elements,
- Ceramic products, etc

## **COMPONENTS OF PREFAB CONSTRUCTION**

- Flooring and Roofing Scheme
- Beams
- Columns
- Walls
- Staircase
- Lintels
- Sunshade/Chajja Projections

# PREFABRICATION SYSTEMS

- The system of prefabricated construction depends on the extent of the use of prefab components, their materials sizes and the technique adopted for their manufacture and use in building.





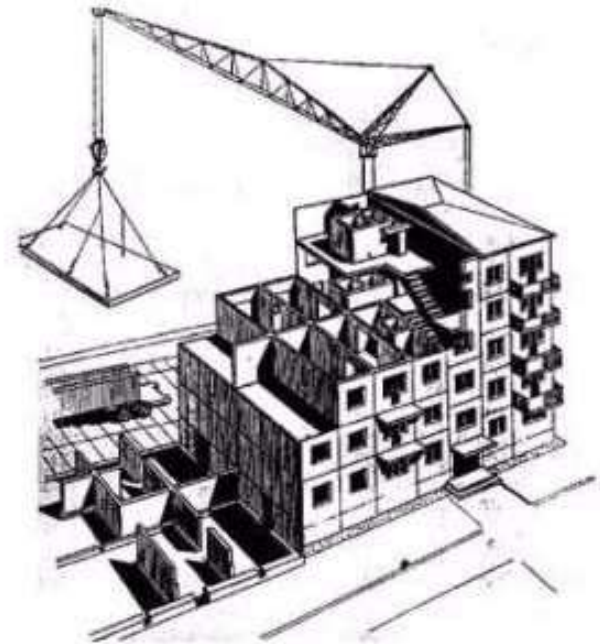
# VARIOUS PREFABRICATION SYSTEMS

## ➤ OPEN PREFAB SYSTEM

- This system is based on the use of the basic structural elements to form whole or part of a building. The standard prefab concrete components which can be used are:-
  - ✓ Reinforced concrete channel units,
  - ✓ Hollow core slabs,
  - ✓ Hollow blocks and battens,
  - ✓ Precast planks and battens,
  - ✓ Precast joists and tiles,
  - ✓ Cellular concrete slabs,
  - ✓ Prestressed/reinforced concrete slabs,
  - ✓ Reinforced/prestressed concrete beams,
  - ✓ Reinforced/prestressed concrete columns,
  - ✓ Precast lintels and chajjas,
  - ✓ Reinforced concrete waffle slabs/shells,
  - ✓ Room size reinforced/prestressed concrete panels,
  - ✓ Reinforced/prestressed concrete walling elements, and
  - ✓ Reinforced/prestressed concrete trusses

## CATEGORIES OF OPEN PREFAB SYSTEMS

- There are two categories of open prefab systems depending on the extent of prefabrication used in the construction as given below:
  - ✓ **Partial prefab open system**
  - ✓ **Full prefab open system**



## **PARTIAL PREFAB OPEN SYSTEM**

- This system basically emphasizes the use of precast roofing and flooring components and other minor elements like lintels, chajjas, kitchen sills in conventional building construction. The structural system could be in the form of in-situ framework or load bearing walls.



## FULL PREFAB OPEN SYSTEM

- In this system almost all the structural components are Prefabricated. The filler walls may be of bricks or any other local material.



## ➤ LARGE PANEL PREFAB SYSTEM

✓ This system is based on the use of large prefab components. The components such as Precast concrete large panels for walls, floors, roofs, balconies, staircase, etc. **The casting of the components could be at the site or off the site.**

MICRO FLATS INSERTED INTO LARGER HOUSING SCHEME



## CLASSIFICATION OF STRUCTURAL SCHEME WITH PRECAST LARGE PANEL

- Wall Systems

- Cross Wall System - In this scheme, the cross wall are load bearing walls whereas the facade Walls are non-load bearing this system is suitable for high rise buildings.
- Longitudinal Wall System - In this scheme the cross walls are non-load bearing whereas Longitudinal walls are load bearing walls. This system is suitable for low rise buildings.
- A combination of the above system with all load bearing walls can also be adopted.



- **PRECAST FLOORS**

- precast flooring units could be homogeneous or non homogeneous.
- ✓ **Homogeneous floors :-** could be solid slabs, cored slabs, ribbed or waffle slabs.
- ✓ **Non-homogeneous floors :-** could be multi-layered ones with combinations of light weight Concrete or reinforced/prestressed concrete, with filler blocks



- **STAIRCASE SYSTEMS**

- Staircase system could consist of single flights with in-built risers and treads in the element only. The flights are normally unidirectional transferring the loads to supporting landing slabs or load bearing walls.

- **BOX TYPE CONSTRUCTION**

- In this system, room size units are prefabricated and erected at site. Toilets and kitchen blocks also be similarly prefabricated and erected at site.
- This system derives its stability and stiffness from the box unit which are formed by the four adjacent walls. Walls are jointed to make rigid connections among themselves. The box unit rests on the plinth foundation which may be of conventional type or precast type.



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Thank you

